

Agenda

- Dimensionality reduction
- Feature selection
- Principle Component Analysis

1. Dimensionality reduction

DIMENSIONALITY REDUCTION

As we have taken so far, Every final decision is dependable on independent variables that we have called features (dimensionality). As time goes, the features started to blow up and increase significantly. The datasets started to be equipped with hundreds of features which raises the complexity of the problem and make it harder to solve.

Problems of high number of features (Curse of dimensionality):

- As number of features increase, number of training data samples will need to increase as well to be able to make a complex model such that you have all possible combinations in your data samples.
- As the number of features increase, the model becomes more complex which can lead to overfitting problem by depending on the the data trained and won't be able to generalize.
- Not all features are important, some features could be misleading that we can't rely on it in our model. However since these features exist in the data, the model will try its best to find a relation due to the no understanding of causality.

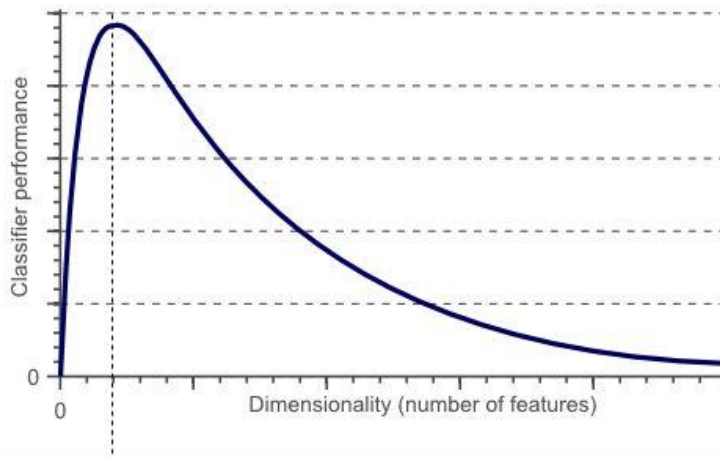
DIMENSIONALITY REDUCTION

- More number of features means more computations which will lead by default to more time and more cost on the hardware needed.
- More number of features will lead by default in an increase in the data storage space needed.
- Some algorithms depend on distance measurements which is less accurate as we go up in dimensional spaces.
- Higher number of features means less visualizations as our minds can't interpret data geometrically unless it is in 1D, 2D or 3D forms which make it impossible to visualize in the hyper space.
- Some features can be highly correlated which we call redundant features (Multicollinearity) that makes our model more prone to errors and noise. For example, storing number of minutes trained on treadmill and calories burnt are useless. As one of them indicates the other.

DIMENSIONALITY REDUCTION

Are less number of features always gives an indication of better model?

No, As number of features increase, the model performance kept going up by gaining more information of the data until we reach an inflection point that more features than this would lead to a consequent decrease in the performance due to the problems we mentioned before.



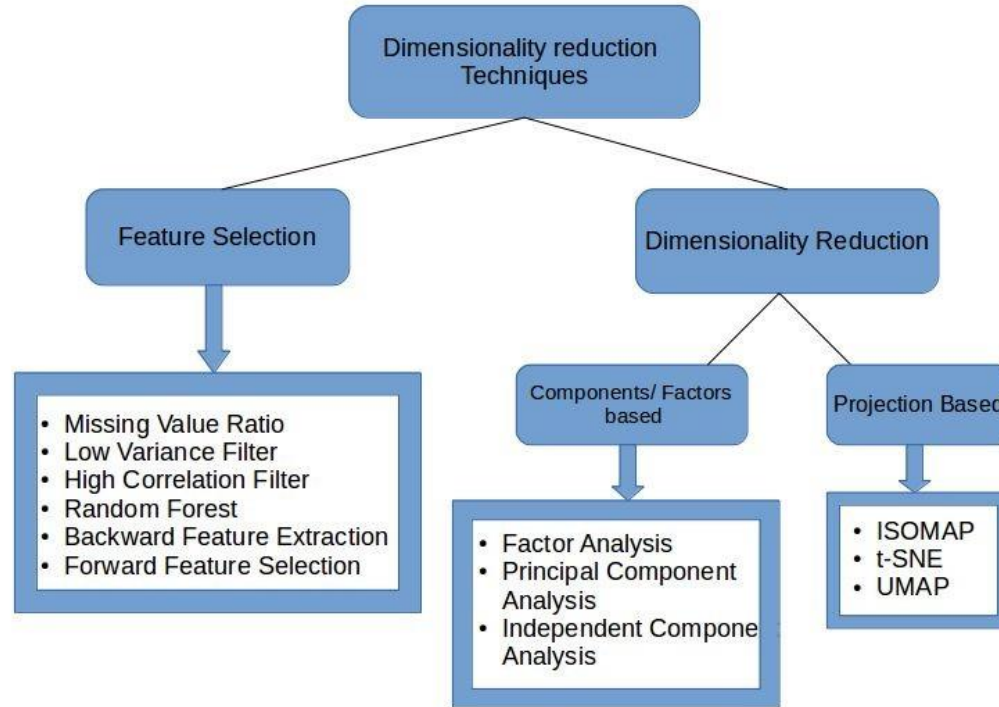
DIMENSIONALITY REDUCTION

Dimensionality reduction methods are techniques which address the curse of dimensionality problem in a way that can enhance interpretability of the model while keeping as much information as we can. Dimensionality reduction techniques are lossy operations as we lose some information that can't be retrieved once again but on the other hand we solved so many problems by such an idea.

Reducing number of features can be categorized into two main categories:

- Feature Selection which is only keeping the most relevant features from the data and remove the redundant ones. Hence no transformations takes place. Techniques is classified into wrapper, filter and embedded methods. (e.g. Random forests, Backward elimination and Forward selection)
- Feature Extraction / Projection Techniques / Dimensionality Reduction / Feature Engineering which is finding smaller set of new variables that are mainly combinations of the input variables. Hence transformations takes place, so the output features are totally different than the input features.

DIMENSIONALITY REDUCTION



FEATURE SELECTION

Let us take some hints about the feature selection methods before digging deep in the feature extraction (Dimensionality reduction) techniques.

There are 3 main classes in the feature selection techniques which are

Filter

A subset of features is selected based on their relationship to the target variable. The selection is not depending on any machine learning algorithms. Filter methods are mainly based on relevance of features using some statistical methods as Pearson's correlation (explained before), Chi-square, ANOVA and Kendall's.

Wrapper

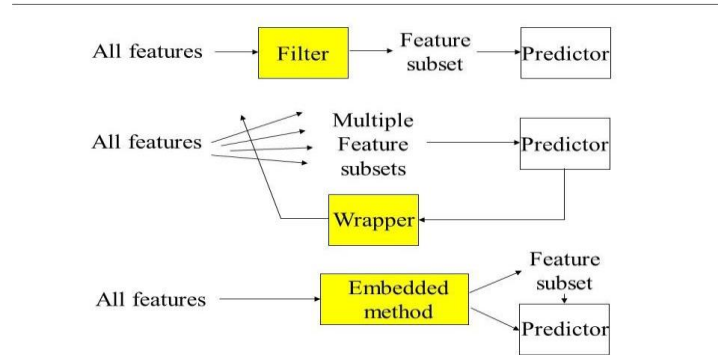
It deals with the problem as a search problem where different combinations are tried out against each other and evaluated to find the best combinations and eliminate the rest. This is like recursive feature elimination algorithm.

2. Feature Selection

FEATURE SELECTION

Embedded

This method learn which features best contribute to the model accuracy while the model is being created. The most known method is the regularization techniques we have taken before as the θ 's values are getting lower that some features are somehow eliminated internally while doing the regression. Types of regression as we took before is the Ridge regression (L2) and LASSO regression (L1). Random forest is among embedded method.



FEATURE SELECTION: FILTER

We have already applied before some feature selection filter methods but we don't know that it was a feature selection filter method. Can you guess what they were?:

- Missing value ratio where we drop the columns that have high number of records with missing values rather than imputing them. This can be assumed as a filtration method using feature selection.
- Low variance filter where we were dropping columns that doesn't have high variance among their values. So if a variable in our dataset having only 1 unique value in it so we drop such value. This can be measured by getting the variance of each column and drop it from your features.
- High Correlation filter where we were plotting the correlation between each variable and another and drop the ones with very high correlation with each other as this means that we have redundant features. This was done by using pearson's correlation.

FEATURE SELECTION: FILTER

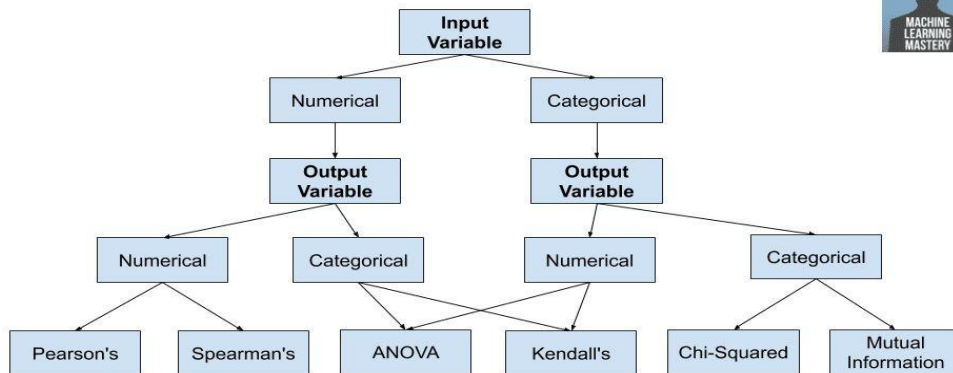
As filter method measures the relevance between each feature and the output variable using statistical methods, let us surf over them quickly to know what each statistical measure is used for.

Pearson's correlation measure the correlation between 2 continuous variables.

CHI-square measure the correlation between 2 categorical variables.

Kendall's and ANOVA measures the correlation between continuous and categorical variables.

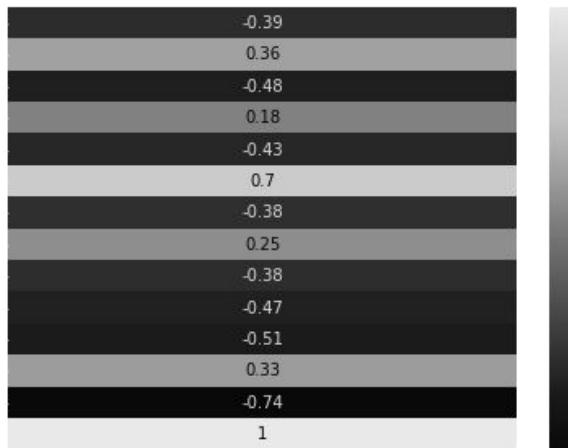
How to Choose a Feature Selection Method



FEATURE SELECTION: FILTER

Pearson's Correlation

A statistical method that measures the linear correlation between two variables which are both continuous. Its value ranges from -1 to 1 where -1 and 1 means means perfect positive or negative linear correlation while 0 means no correlations. What we do here is the getting the correlation between each variable and the target variable:



FEATURE SELECTION: FILTER

Pearson's Correlation

By putting the threshold of 0.5 for example, we can find that only 3 features of the 13 features have correlation of value higher than 0.5 or lower than -0.5 which are RM, PTRATIO and LSTAT. Note that this shows correlation between each feature and the output variable. However we should check for correlations between the 3 variables themselves to avoid redundant features. The check shows the following:

LSTAT and RM are highly correlated so we can remove one of them as well.



FEATURE SELECTION: FILTER

CHI-squared

As we said so far that we are aiming to find the features which are highly correlated with the target variable as the previous example but now with categorical features. If two variables are independent hence the chi-squared value will be low and vice versa.

Formula:

The Formula for Chi Square Is

$$\chi_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where:

c = degrees of freedom

O = observed value(s)

E = expected value(s)

Steps:

- 1 Define hypothesis
- 2 Build contingency table (Confusion Matrix)
- 3 Find expected value
- 4 Calculate Chi-square value
- 5 Accept or reject the hypothesis

FEATURE SELECTION: FILTER

Consider a data-set where we have to determine why customers are leaving the bank, let's perform a Chi-Square test for two variables. Gender of a customer with values as Male/Female as the predictor and Exited describes whether a customer is leaving the bank with values Yes/No as the response. In this test we will check is there any relationship between Gender and Exited. Let us perform Chi-squared test.

Build the hypothesis

Null hypothesis $\Rightarrow$ The two variables are independent. Alternate

hypothesis $\Rightarrow$ The two variables are dependent. Build Contingency table

Exited\ Gender	Yes	No	Total
Male	38	178	216
Female	44	140	184
Total	82	318	400

FEATURE SELECTION: FILTER

The degree of freedom of this contingency table is $(r-1) \times (c-1)$ where r and c are the number of rows and number of columns in the contingency table respectively.

Hence degree of freedom (df) is $(2-1) \times (2-1) = 1$. Find the expected value

Based on null hypothesis that two variables are independent: Hence the expected value given male and exited is as follows:

$$P(A \cap B) = P(A) * P(B)$$

$$E1 = n * p$$

$$p = p(\text{Yes}) * p(\text{Male})$$

$$p = (82/400) * (216/400)$$

$$p = 0.1107$$

$$\text{now, } E1 = 400 * 0.1107 = 44$$

FEATURE SELECTION: FILTER

Similarly calculate for all possible cases (female & exited)(male & not exited)(female & not exited).
So we get the following results:

Exited\Gender	Yes	No
Male	44	172
Female	38	146

The Formula for Chi Square Is

$$\chi_c^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

where:

c = degrees of freedom

O = observed value(s)

E = expected value(s)

Calculate CHI-squared value based on its equation: $\Rightarrow$

The result is as follows that chi-squared is 2.22.

Gender,Exited	O	E	O-E	Square of O-E	(Square of O-E) / E
Male,Yes	38	44	-6	36	0.818181818
Male,No	178	172	6	36	0.209302326
Female,Yes	44	38	6	36	0.947368421
Femal,No	140	146	-6	36	0.246575342
Chi Square Value					2.221427907

FEATURE SELECTION: FILTER

Here comes the question, is this value high or low? Accept or reject null hypothesis?

Chi-square has a table that defines if this chi-squared value is good enough to assume dependence or not based on critical value. The critical value is defined by the degree of freedom (df) and the level of confidence.

The level of confidence defines the critical value that if the chi-square value falls below, we accept the null hypothesis. α is the complement of confidence value where if we say α of 0.1 hence we mean confidence level of 90% or 0.9. As we go higher in α , the area to the right of the critical value is getting larger which means more loose feature selection. Hence if α is 1, all variables are depending on the target variable hence no feature selection is done.

In our example assuming a 95% confidence where we want to perform a very hard feature selection, the critical value is 0.05 hence the critical value to compare is 3.84 which means that chi-squared value of 2.22 is less than it so we accept the null hypothesis hence the two variables are independent and we discard this feature.

FEATURE SELECTION: FILTER

Degrees of Freedom	Chi-Square (χ^2) Distribution									
	Area to the Right of Critical Value									
	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	—	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.299
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.265	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.542	10.982	12.338	14.042	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.844	15.379	17.292	35.563	38.885	41.923	45.642	48.290
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963	49.645
28	12.461	13.565	15.308	16.928	18.939	37.916	41.337	44.461	48.278	50.993
29	13.121	14.257	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.954	16.791	18.493	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.707	32.357	34.764	37.689	63.167	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.739	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.196	61.754	65.647	69.126	73.291	107.565	113.145	118.136	124.116	128.299
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

FEATURE SELECTION: WRAPPER

As we have said before, wrapper method acts as a searching problem where we try out different combinations using two methods which are backward feature elimination and forward feature elimination.

Backward Feature Elimination:

- We first take all the n variables present in our dataset and train the model using them.
- We then calculate the performance of the model.
- Now, we compute the performance of the model after eliminating each variable (n times), so we drop one variable every time and train the model on the remaining $n-1$ variables.
- We identify the variable whose removal has produced the smallest (or no) change in the performance of the model, and then drop that variable
- Repeat this process until no variable can be dropped.

FEATURE SELECTION: EMBEDDED

For Embedded method, Just revise for the regularization part and the random forest algorithm.

Since both wrappers and embedded are using machine learning algorithms, so what is the difference between them?

Both wrapper and embedded methods are similar in using machine learning algorithms but have some difference in that Wrapper methods consider unimportant features iteratively based on the evaluation metric, while Embedded methods perform feature selection and training of the algorithm in parallel. In other words, the feature selection process in the embedded method is an integral part of the classification/regressor model. This is the type that random forest is a part of it as it performs LASSO method internally.

FEATURE EXTRACTION

Feature extraction is a type of dimensionality reduction where transformations take place as the input variables are combined together in ways to produce new variables that can retain most of the information in the original input variables. This can be categorized to:

- Linear Methods (Factors-based) as principal component analysis (PCA), Factor analysis (FA) and Independent component analysis (ICA)
- Non-Linear Methods (Projection-based or Manifold Learning) as Kernel PCA, t-distributed Stochastic Neighbor Embedding (tSNE), Multidimensional Scaling (MDS), [Kohonen Self-Organizing Map \(SOM\)](#), and Uniform manifold approximation and projection (UMAP).

Our course will cover Principal Component Analysis (PCA) in addition to the Self-Organizing Map (SOM) in the artificial neural network part that we will take later.

3. Principle Component Analysis

PRINCIPAL COMPONENT ANALYSIS (PCA)

PCA is the most popular dimensionality reduction method used which is by default an unsupervised technique as it doesn't need any target labels. PCA gets some **linearly uncorrelated and orthogonal components called principal components such that each component is a linear combination of some features of the original data where those uncorrelated components preserve as much as possible from the total variance of the data** noting that variance means how much the data is spread around the mean.

How principal components are organized?

Principal components are organized such that the more early component in the data, the more variance is retained. As we go further in the principal components less variance is retained. This means that the first component in the principal components carries the most information (explain the most variance) inside it. Then the second principal component try to explain the variance of the original data set that the first component couldn't capture then the third component tries to explain the variance of the dataset that the first 2 components couldn't capture and so on .

PRINCIPAL COMPONENT ANALYSIS (PCA)

Equations of the principal components:

$Z_1 = 1\text{st principal component} = \Phi_{11} X_1 + \Phi_{21} X_2 + \dots + \Phi_{n1} X_n$

Similarly:

$Z_2 = 2\text{nd principal component} = \Phi_{12} X_1 + \Phi_{22} X_2 + \dots + \Phi_{n2} X_n$ and

so on with all the principal components.

What are the pre-conditions to apply PCA?

- Features should be scaled (preferably standardized) as if we don't scale them, the variables of the largest scale will inherit inside them the largest variance so the loading of the coefficients will be biased to them.
- PCA is applied to numerical data so categorical data is not included. To be able to deal with categorical data, we should apply optimal scaling where it tries to find out the best values to assign to each category to make distance metric best.

REMEMBER: VARIANCE

Standard deviation gives an indication on how far a group of numbers is from the mean of this group. Standard deviation is the square root of the variance.

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

Variance is the average of the squared differences from the mean.

Both of them are Indicating the spread of the data with more stressing of variance on the difference by squaring it to give an indication of extreme outliers.

REMEMBER: VARIANCE

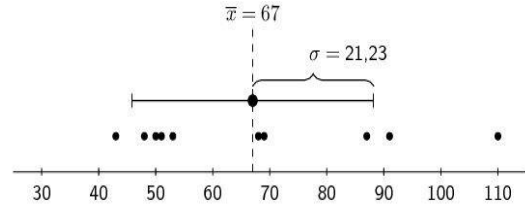
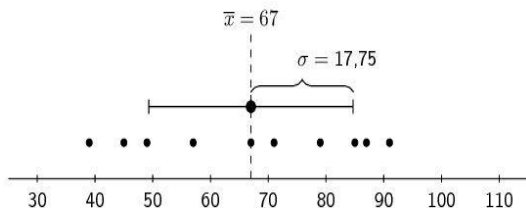
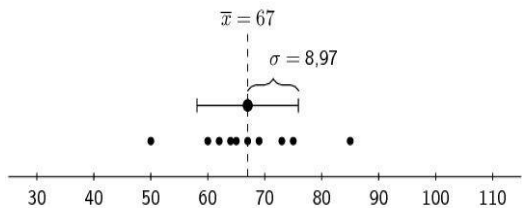
Example: Assume having the following 3 datasets D1 =

{65;75;73;50;60;64;69;62;67;85}

D2 = {85;79;57;39;45;71;67;87;91;49}

D3 = {43;51;53;110;50;48;87;69;68;91}

The 3 datasets are sharing the same mean with a value of 67. However, each one has a totally different standard deviation of values 8.97, 17.75 and 21.23 respectively.



REMEMBER: COVARIANCE

A measure of the direction of the correlation between two or more sets of random variables such that if covariance of x, y is positive then x and y are in same direction so if x increase, y increases and vice versa. If covariance of x, y is negative then x, y are in opposite direction then if x increases, y decreases and if covariance is 0 then x and y are uncorrelated. It's actually the sign of the covariance that matters.

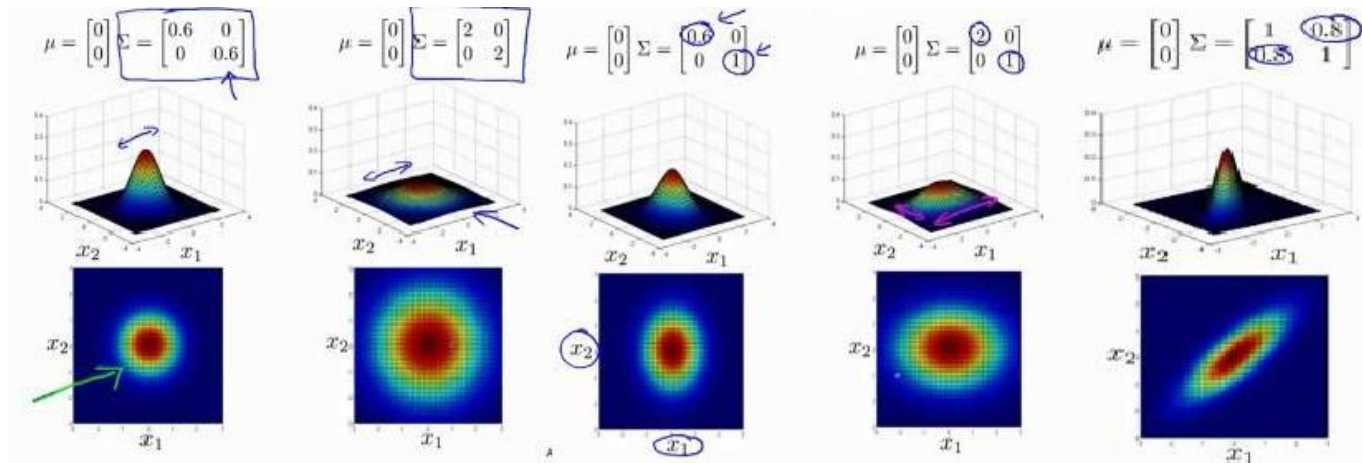
When we say n×n covariance matrix this means that the diagonal contains the variance of each feature (feature1, feature2, ..., feature n) and the off-diagonal values are the covariance between the i,j of this place. so covariance matrix is symmetric.

$$var(x) = \frac{\sum (x_i - \bar{x})^2}{N}$$

$$cov(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

REMEMBER: COVARIANCE

$$\begin{bmatrix} \text{Var}[Y_1] & \text{Cov}[Y_1, Y_2] & \text{Cov}[Y_1, Y_3] & \cdots & \text{Cov}[Y_1, Y_n] \\ \text{Cov}[Y_2, Y_1] & \text{Var}[Y_2] & \text{Cov}[Y_2, Y_3] & \cdots & \text{Cov}[Y_2, Y_n] \\ \text{Cov}[Y_3, Y_1] & \text{Cov}[Y_3, Y_2] & \text{Var}[Y_3] & \cdots & \text{Cov}[Y_3, Y_n] \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \text{Cov}[Y_n, Y_1] & \text{Cov}[Y_n, Y_2] & \text{Cov}[Y_n, Y_3] & \cdots & \text{Var}[Y_n] \end{bmatrix}$$



LINEAR TRANSFORMATION

A linear transformation is a function from one vector space to another that respects the underlying (linear) structure of each vector space. A linear transformation is also known as a linear operator or map.

The defining characteristic of a linear transformation $T : V \rightarrow W$ is that, for any vectors v_1 and v_2 in V and scalars a and b of the underlying field,

$$T(a v_1 + b v_2) = a T(v_1) + b T(v_2)$$

Imagine we have a square. The square, like your hand forming an “L”, has a particular orientation which we graphically represent with arrows:



LINEAR TRANSFORMATION

We can manipulate this square in certain ways. For example, we can pull or push on an edge to stretch or compress the square (Figure A and B). We can rotate the square (Figure C) or flip it to change its orientation—imagine turning your hand over so that the “L” becomes an “J” (Figure D). We can even shear the square, which means to deform it by applying a force either up, down, left, or right at one of the square’s corners (Figure E). Intuitively, a linear transformation is one in which a straight line before the transformation results in a straight line after the transformation



MATRIX RANK

Matrix rank is the number of linearly independent rows or columns in the matrix. Assume the following matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \end{bmatrix}$$

Although this matrix is of 2 rows but its rank is 1 as the second row is 3 x first row.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 1 & 4 & 5 \end{bmatrix}$$

Here the second row is not dependable on 1st row or 3rd row, however the 3rd row is the summation of both 1st and 2nd row hence dependable. The rank here is 2.

MATRIX RANK

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 1 & -2 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

What is the matrix rank of the above matrices?

For the first matrix, the 3rd row is the 1st row - 2 x 2nd row which means that it is dependable. Hence the matrix rank is 2.

For the second matrix, it is obvious that there is no relation between its rows or columns so the matrix rank is 3.

Note: The only matrix with rank 0 is the null matrix (matrix with all zeros).

EIGEN VALUES AND EIGEN VECTORS

If you can draw a line through the three points $(0,0)$, v and Av , then Av is just v multiplied by a number λ ; that is, $Av=\lambda v$. In this case, we call λ an eigenvalue and v an eigenvector. For example:

$$Av = \begin{pmatrix} 1 & 2 \\ 8 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \lambda v.$$

Hence $(1,2)$ is an eigenvector and 5 is an eigenvalue.

SINGULAR VALUE DECOMPOSITION

The intuition behind this is that every matrix has a close matrix of lower rank to it that can act as the best approximation to the original higher rank matrix. Note that SVD can be applied to all matrices (rectangular or square) which is a very good benefit. SVD of say matrix A means the factorization of A to 3 matrices U , D & V where columns of U and V are orthonormal (orthogonal and normalized) and D is diagonal with positive entries (Covariance) $\Rightarrow A = U \cdot D \cdot V^T$ or $A = U \cdot \Sigma \cdot V^T$

U consists of the eigenvectors of $A \cdot A^T$, while V is a matrix of eigenvectors of $A^T \cdot A$. The singular values in Σ are the square root of the eigenvalues of $A \cdot A^T$.

Note: Assuming A is of dimension $n \times p$, hence U is of dimension $n \times p$, Σ is of dimension $p \times p$ while V is of dimension $p \times p$

PRINCIPAL COMPONENT ANALYSIS (PCA)

Algorithm steps:

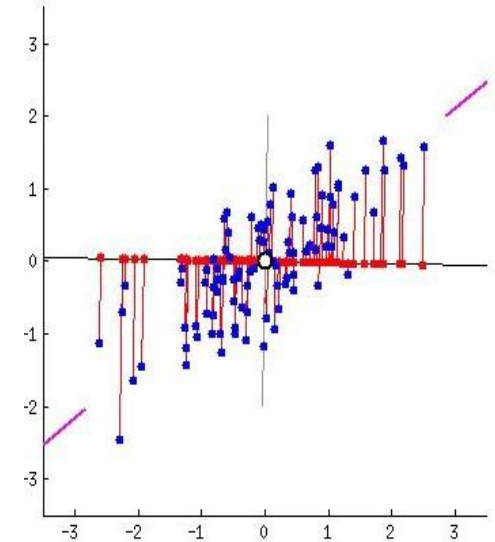
- 1- Standardize the data
- 2- Calculate the covariance matrix of these data. (Σ)
- 3- Calculate the eigenvectors of this covariance and sort it according to their eigenvalues.
- 4-Take the first K columns of the eigenvector (Feature vector).
- 5- Multiply this matrix by the original data.

To check that these PC are right just calculate the covariance matrix of this new formed reduced matrix and you will see that the diagonal values are non-zeros and all off-diagonal values are zeros which means that all the principal components are uncorrelated. BUT, maybe we have some questions about step 3 and 4 which are why to calculate the eigenvectors of the covariance matrix and why taking first K components works as PC?

PRINCIPAL COMPONENT ANALYSIS (PCA)

Eigenvectors and eigenvalues are the linear algebra concepts that we need to compute from the covariance matrix in order to determine the principal components of the data.

As there are as many principal components as there are variables in the data, principal components are constructed in such a manner that the first principal component accounts for the **largest possible variance** in the data set. For example, let's assume that the scatter plot of our data set is as shown below, can we guess the first principal component? Yes, it's approximately the line that matches the purple marks because it goes through the origin and it's the line in which the projection of the points (red dots) is the most spread out. Or mathematically speaking, it's the line that maximizes the variance (the average of the squared distances from the projected points (red dots) to the origin).

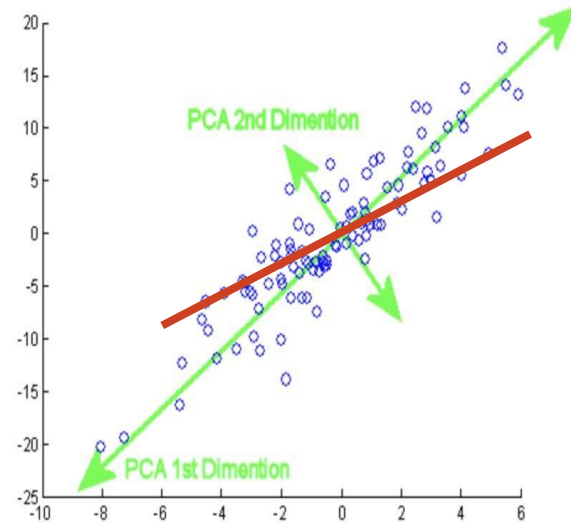
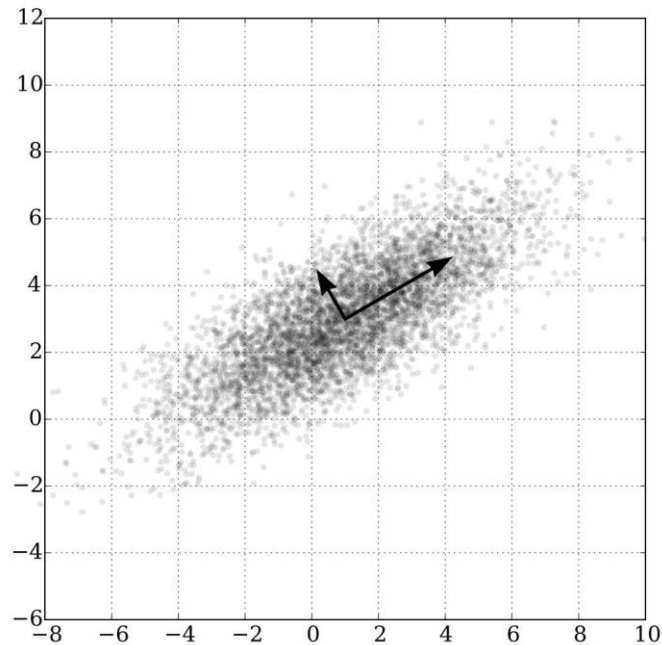


PRINCIPAL COMPONENT ANALYSIS (PCA)

The second principal component is calculated in the same way, with the condition that it is uncorrelated with (i.e., perpendicular to) the first principal component and that it accounts for the next highest variance. This continues until a total of p principal components have been calculated, equal to the original number of variables.

Now that we understood what we mean by principal components, let's go back to eigenvectors and eigenvalues. What you firstly need to know about them is that they always come in pairs, so that every eigenvector has an eigenvalue. And their number is equal to the number of dimensions of the data. For example, for a 3-dimensional data set, therefore there are 3 eigenvectors with 3 corresponding eigenvalues. Hence eigenvectors and eigenvalues who are behind all the magic explained above, because the eigenvectors of the Covariance matrix are actually the directions of the axes where there is the most variance (most information) and that we call Principal Components. And eigenvalues are simply the coefficients attached to eigenvectors which by ranking your eigenvectors in order of their eigenvalues, highest to lowest, you get the principal components in order of significance.

PRINCIPAL COMPONENT ANALYSIS (PCA)



PRINCIPAL COMPONENT ANALYSIS (PCA)

Can we consider the red line in the previous image a 3rd principal component?

No we can't as the principal components should be orthogonal (perpendicular) to each other. Hence the 3rd dimension to these 2 principal components will be going in or out of the screen.

As we saw in the previous step, computing the eigenvectors and ordering them by their eigenvalues in descending order, allow us to find the principal components in order of significance. In this step, what we do is, to choose whether to keep all these components or discard those of lesser significance (of low eigenvalues), and form with the remaining ones a matrix of vectors that we call Feature vector.

So, the feature vector is simply a matrix that has as columns the eigenvectors of the components that we decide to keep. This makes it the first step towards dimensionality reduction, because if we choose to keep only p eigenvectors (components) out of n , the final data set will have only p dimensions.

PRINCIPAL COMPONENT ANALYSIS (PCA)

In this step, which is the last one, the aim is to use the feature vector formed using the eigenvectors of the covariance matrix, to reorient the data from the original axes to the ones represented by the principal components (hence the name Principal Components Analysis). This can be done by multiplying the transpose of the original data set by the transpose of the feature vector.

$$FinalDataSet = FeatureVector^T * StandardizedOriginalDataSet^T$$

PRINCIPAL COMPONENT ANALYSIS

What should be the number of components K that we choose?

Instead of arbitrarily choosing the number of dimensions to reduce down to, it is simpler to choose the number of dimensions that add up to a sufficiently large portion of the variance (e.g., 95%).

Unless, of course, you are reducing dimensionality for data visualization—in that case you will want to reduce the dimensionality down to 2 or 3.

Yet another option is to plot the explained variance as a function of the number of dimensions. There will usually be an elbow in the curve, where the explained variance stops growing fast. In this case, you can see that reducing the dimensionality down to about 100 dimensions wouldn't lose too much explained variance.

